

The Most Effective Wrestling Gameplan with the Help of Data Analysis

Applying Multivariate Statistics to World and Continental
Championship Matches



Department of Mathematics

National and Kapodistrian University of Athens

Multivariate Data Analysis

Chatzipapas Tilemachos

A.M.:1112202200165

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Abstract: Wrestling is an ancient, beautiful, and incredibly complex sport. Beginners are often overwhelmed by the sheer volume of available techniques. This study uses data analysis to cut through the noise and build the most effective, data-driven starting gameplan. By analyzing 36 elite matches from UWW events using Principal Component Analysis (PCA), Factor Analysis (FA), and Linear Discriminant Analysis (LDA), we shift the focus from "scoring flashy points" to "actually winning the match". Our results reveal a concrete baseline strategy that every wrestler can adopt as an optimal starting point before customizing their personal style.

Keywords: Freestyle Wrestling, Sports Analytics, Principal Component Analysis, Factor Analysis, Linear Discriminant Analysis, Combat Sports Strategy.

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1. Introduction

Wrestling is an old sport and it has evolved drastically over the years. Today, there are hundreds of moves, and for every single move, there is always a counter-move, and a counter for that counter. For a new athlete starting out, this universe of techniques is completely overwhelming. How can you realistically improve if you are trying to practice 20 different moves at once? How many moves should you actually focus on, and which ones should you pick?

In this study, we deliberately choose not to look for the moves that score the "most points". Why? Because at the end of the day, we don't care about scoring point-tallies; we care about winning the match. You might ask: "Aren't those the same thing?" No, they are not. If you hit a high-flying 2-point move that leaves you completely exhausted, causing you to give up 4 points right after because you gassed out, then that move is a bad investment. We only care about how points scored in any way (full execution, push-outs, or passivity) statistically translate into a Win.

2. Materials

The dataset for this study was collected by human eye tracking, watching tape from various Continental and World Championships. Every single match was sourced from the official United World Wrestling (UWW) YouTube channel.

Sample Size: The final database consists of 36 elite matches featuring roughly 66 different world-class athletes.

Variables: For each match, 17 technical and tactical variables were recorded as columns, representing the points scored via specific actions (e.g., Attack, Counter-Attack, Low-Single, Double-Leg, Go-Behind, Duck-Under, Upper-Body-Throws, Gut-Wrench, Leg-Lace, Opponent-Inactivity, Points-From-Pushing-Out-Of-Bounds, etc.).

Target Group: A binary variable indicating the match outcome (1 for Win, 0 for Loss).

3. Methods

Note for the Wrestling Community: If math bores you, feel free to skip this section and the next one! Full code scripts and raw data will be on my GitHub repository (link in the Appendix).

For the academic evaluation of this paper, the following multivariate techniques were implemented via R programming:

* 3.1 Principal Component Analysis (PCA)

PCA was used to examine the dimensionality of wrestling. By finding the eigenvalues λ_j of the Correlation Matrix R , we analyze how much variance each component captures. The total variance is preserved through the trace identity:

$$\sum_{j=1}^p s^2(Z_j) = \text{tr}(R) = p$$

We retain components with eigenvalues $\lambda_j \geq 1$ (*Kaiser's Criterion*) to see how many dimensions are truly required to describe an elite match.

* 3.2 Factor Analysis (FA)

To uncover underlying playstyles, we fit an orthogonal factor model:

$$X - \mu = LF + U$$

where L represents the factor loadings, F the common latent factors, and U the unique variance matrix Ψ . The covariance matrix is decomposed as $\Sigma = LL' + \Psi$.

To achieve Simple Structure and make the playstyles interpretable, we applied Varimax Rotation, maximizing the variance of the squared loadings across the factors:

$$\max_T \sum_{k=1}^m \left[\frac{1}{p} \sum_{j=1}^p (l_{jk}^*)^4 - \left(\frac{1}{p} \sum_{j=1}^p (l_{jk}^*)^2 \right)^2 \right]$$

* 3.3 Linear Discriminant Analysis (LDA)

To find out what actually drives a victory, we used Fisher's LDA. We decompose the total sample scatter matrix into Within-group (W) and Between-group (B) matrices ($T = W + B$).

We maximize Fisher's criterion to find the linear combination

$$Y = u'X$$

that offers the greatest separation between Wins (1) and Losses (0):

$$\max_u \lambda(u) = \frac{u'Bu}{u'Wu} \implies W^{-1}Bu = \lambda u$$

The resulting coefficients (LD1) provide the mathematical weights of each wrestling move regarding its contribution to winning.

4. Results

4.1 Exploratory Data Analysis & Technical Action

Profiles

To establish a baseline understanding of how scoring occurs in freestyle wrestling matches, we first examine which technical actions are executed most frequently and which ones contribute the most to the scoreboard. Figure 1 illustrates a side-by-side comparison of the total frequency of each action against the total points it generated across the entire dataset.

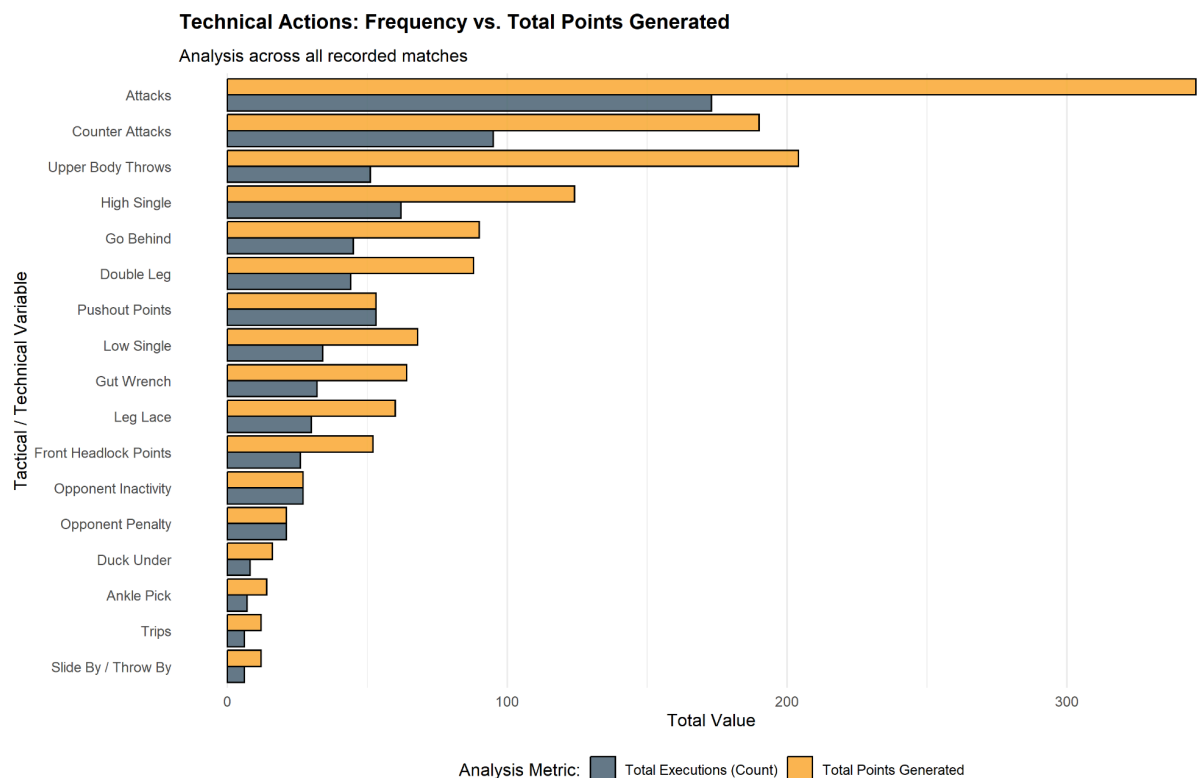


Figure 1

As shown in *Figure 1*, there is a clear distinction between high-frequency setups and high-yield scoring actions. General Attacks and Counter Attacks occur at the highest frequencies, serving as the primary drivers of match activity. However, when evaluating actual point generation, heavy scoring actions like Upper Body Throws and ground-work techniques (Gut Wrench and Leg Lace) show an exponential spike in point yield relative to their execution frequency.

This indicates that while neutral setups and single-leg entries dictate the pace of the match, parterre dominance and high-amplitude throws are the most efficient methods for securing large point margins. Critically, the lower frequency of these high-yield actions reflects the tactical reality of combat: parterre scoring carries a strict positional prerequisite, requiring a prior successful takedown to be accessed, while upper-body throws involve high technical risk and exposing one's own hips, making them far harder and more dangerous to execute in live competition.

4.2 Multivariate Statistical Outputs

Running the customized R script on our clean dataset yielded the following distinct outputs:

4.2.1 PCA Dimension Check

The eigenvalues of our correlation matrix did not drop sharply. In fact, we need *8 different Principal Components* to satisfy *Kaiser's criterion* ($\lambda \geq 1$), with the first two explaining *only 12.91% and 11.79%* of the variance respectively.

4.2.2 FA Playstyle Archetypes

The Varimax rotation successfully achieved an excellent Simple Structure, highlighting two primary hidden playstyles (Factors):

- Factor 1 (The Aggressive Initiator):

Heavy loading on general Attack (0.992)
and standard setups like Double Leg (0.433).

- Factor 2 (The Tactical Counter-Wrestler):

Dominant loading on Go Behind (0.997)
and Counter Attack (0.557).

4.2.3 LDA Winning Weights (LD1)

The Linear Discriminant function revealed the coefficients that **separate moves that yield points from those that secure a win:**

- Opponent Inactivity: +1.326

- Pushing Out Of Bounds: +0.484

- Low Single: +0.471

- Duck Under: +0.355
- Leg Lace: +0.330
- Slide By And Throw By: +0.314
- Upper Body Throws: +0.292
- Counter Attack: +0.229
- Attack: +0.198
- High Single: +0.159
- Gut Wrench: +0.151
- Double Leg: +0.143
- Trips: +0.135
- Ankle Pick: +0.054
- Go Behind: +0.019
- Points From Front Head Lock: -0.214
- Opponent Penalty: -0.297

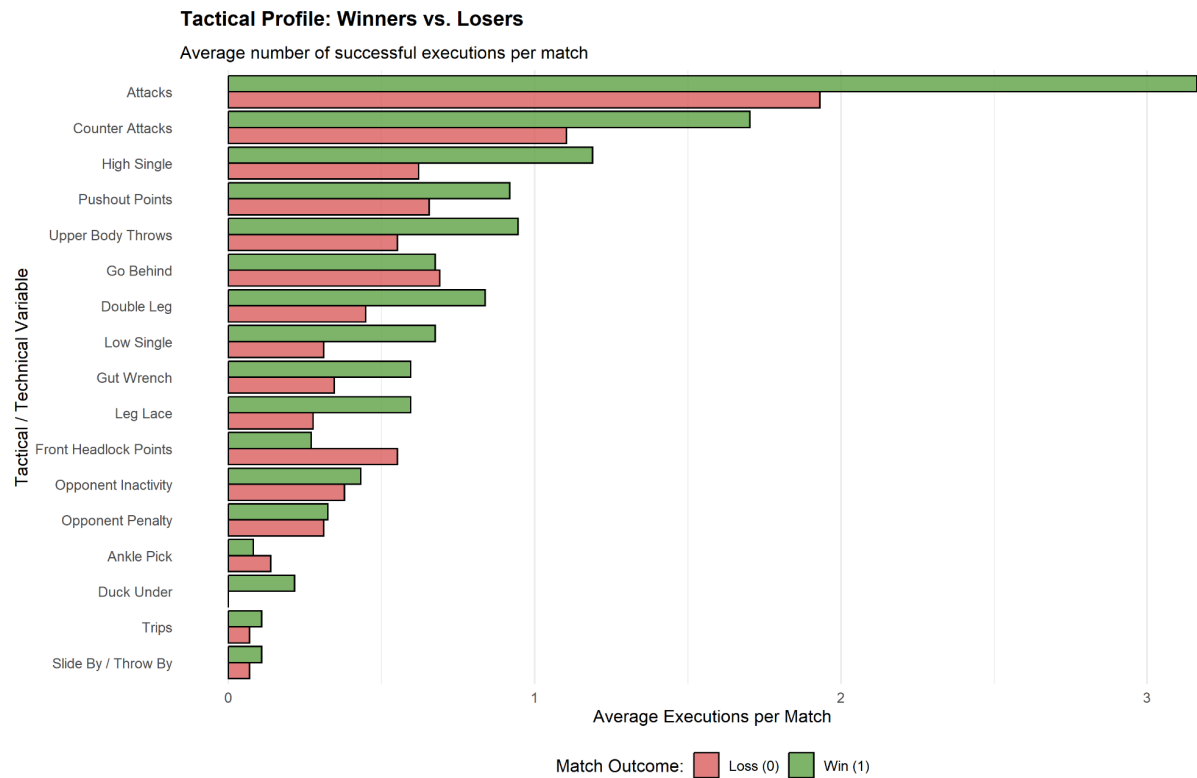


Figure 2

5. Discussion

5.1 Deconstructing the LDA Model: The Inactivity Paradox

Before analyzing the most effective scoring actions, it is crucial to address a specific structural anomaly within the predictive model. A closer inspection of the LDA results reveals a finding that may initially seem counterintuitive when compared directly to the descriptive data. In *Figure 2*, the average points obtained from Opponent Inactivity appear nearly identical for both winners and losers (hovering around 0.4 executions per match).

Yet, the LDA algorithm ranks Opponent Inactivity as the single most critical variable for separating the two groups, while ranking Attacks, which shows a massive visual gap between winners and losers, much lower.

This apparent paradox is explained by how multivariate models evaluate variables. There are two core statistical reasons for this behavior:

The Multicollinearity Effect (Shared Information):

In wrestling, an "Attack" does not happen in a vacuum; it is heavily correlated with specific finishes like a High Single, Double Leg, or Go Behind. Because these variables share the same statistical space, the DA model has to split the credit among them. The predictive power of the Attack variable is essentially diluted across all specific takedowns. Conversely, Opponent Inactivity is a completely isolated variable. It represents a tactical phase that is entirely uncorrelated with your offensive shooting style, meaning it brings 100% unique, undiluted information to the mathematical model.

The Variance and Overlap Factor:

While the green and red bars for Attacks look far apart, the actual data points are highly volatile, ranging anywhere from 0 to 8 attacks per match. This creates a massive statistical overlap between the two groups. Opponent Inactivity, on the other hand, has incredibly low variance, it is almost strictly a binary 0 or 1. Because its distribution is so tightly controlled, even the slightest, most consistent systematic difference in how those 1s are distributed is treated by the DA model as a highly reliable, rock-solid anchor for classification.

Therefore, while Attacks show the largest volume difference, Opponent Inactivity provides the cleanest, most unpolluted mathematical boundary for group separation.

5.2 Interpretation of Multivariate Outputs: PCA, FA, and DA

5.2.1 Wrestling is Beautifully Complex (PCA Interpretation)

While it is clear that each technical action and scoring method carries distinct tactical value, PCA mathematically proves the high dimensionality and complexity of wrestling data. The fact that the variance is distributed across multiple orthogonal dimensions confirms that performance cannot be captured by a single dominant style.

Because we need 8 distinct components to capture the baseline information, we prove that wrestling cannot be reduced to 2 or 3 cheap variables. You cannot describe a Low Single using a Double Leg; every technique occupies its own unique mathematical dimension in elite combat.

5.2.2 The Two Playstyles (Factor Analysis Interpretation)

Our factor model mapped out two beautifully logical blueprints for wrestlers:

1) Style 1 is the aggressive proactive wrestler who forces the action with direct attacks.

2) Style 2 is the heavy-pressure counter-wrestler. Defending doesn't mean you aren't wrestling; it means you are forcing handfights, maintaining position, and waiting for the opponent to make a mistake, punishing them instantly with a Go Behind or a Body Throw.

5.2.3 What ACTUALLY Wins Matches? (LDA)

The weights of LD1 provide a solid, low-risk blueprint for success:

Pace and Pressure Control the Mat: The absolute highest weights belong to Opponent Inactivity (+1.326) and Pushing Out Of Bounds (+0.484). Old coaches always say, "He who controls the handfights and the center wins the match," and the data backs them up 100%. If you dictate the pace, your opponent either gets a passivity warning, giving you the "sweetest", lowest-risk point in wrestling, or they run out of bounds giving you step-out points.

Low-Risk Top-Tier Attacks: When it comes to taking people down, Low Single (+0.471) and Duck Under (+0.355) are massively rewarding. They are low-risk moves that don't require you to overextend or risk getting countered as much as with Body Throws / Double Legs etc.

Finish on the Mat: On the ground, the Leg Lace (+0.330) is twice as effective at sealing a victory compared to the Gut Wrench (+0.151).

The Traps to Avoid: Notice that Points From Front Head Lock turned out negative (-0.214). While the front headlock is a great control position, stalling there or failing to finish the score actively backfires, opening you up to scrambles or stalling calls.

5.3 How to Use This to Improve

If you are a beginner or intermediate wrestler overwhelmed by choices, start here.

Don't practice 20 random moves. Build a playstyle around heavy center-mat pressure and moves that work on the mat. This is your statistically optimized baseline.

Once you dominate with this foundation, you can slowly deviate and add flashier, high-risk moves that suit your personal creativity. Elite athletes execute whatever feels right in the moment without thinking, but everyone needs a smart place to start.

5.4 Strategic Technical Chains

Before addressing the limitations of this study, it is highly practical to examine how specific techniques group together in real combat. While individual move frequencies tell us what scores, the Linear Correlation Heatmap (*Figure 3*) reveals how these scores chain together, providing a technical blueprint for elite performance.

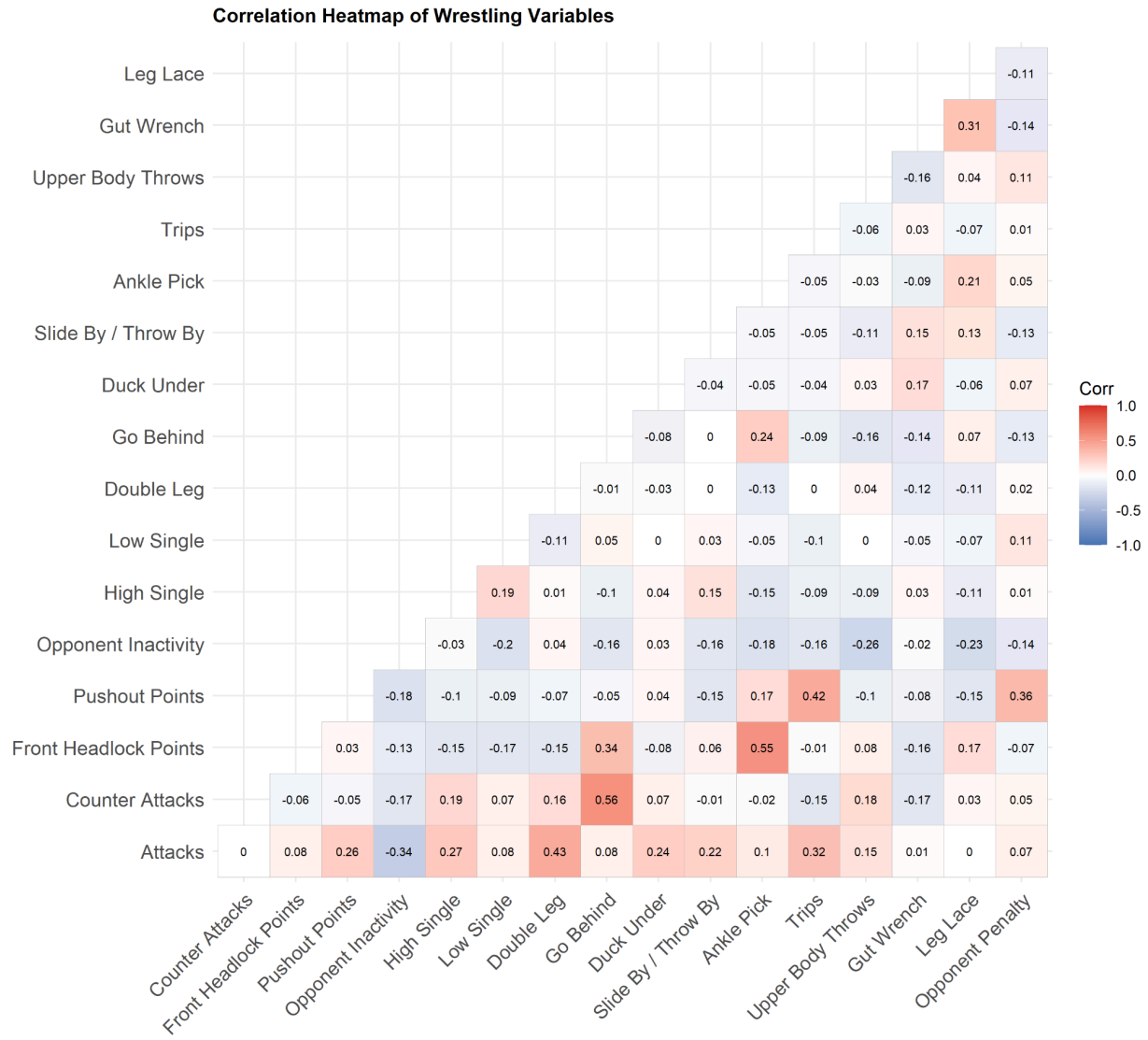


Figure 3

Wrestlers can utilize this heatmap as a diagnostic tool to build highly logical tactical systems. Instead of selecting moves at random, an athlete can look at their preferred primary attack and immediately identify which secondary techniques complement it based on empirical match data.

For example, the heatmap reveals a strong, distinct correlation between Front Headlock Points and the Ankle Pick. In practice, this tactical chain makes perfect competitive sense: when a wrestler secures a tight front headlock and threatens a heavy snap-down, the opponent naturally panics about being pulled down to the mat. To counter this downward pressure, the opponent is forced to post their weight forward and step their feet closer to maintain balance. This defensive reaction inadvertently brings their lead leg within structural reach, perfectly exposing them to a devastating ankle pick. By identifying these empirical clusters, coaches and elite athletes can optimize training time, focusing specifically on technical chains that naturally feed into one another during high-level competition.

5.5 Study Limitations

No data model is perfect. High-amplitude actions like Upper Body Throws (+0.292) look solid here because elite wrestlers score big when they hit them. However, our current database cannot track the failed attempts, the throws that missed and handed the opponent an easy counter-attack score. Additionally, a 36-match dataset is an exceptional diagnostic sample, but expanding it further will only make these insights more reflective of absolute reality.

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* **Appendix**

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The full database, the R scripts that ran the PCA, FA, and DA models, and the code for all the graphs are all on my GitHub:

GitHub Repository:

https://github.com/Tilemachoc/Wrestling_Multivariate_Analysis